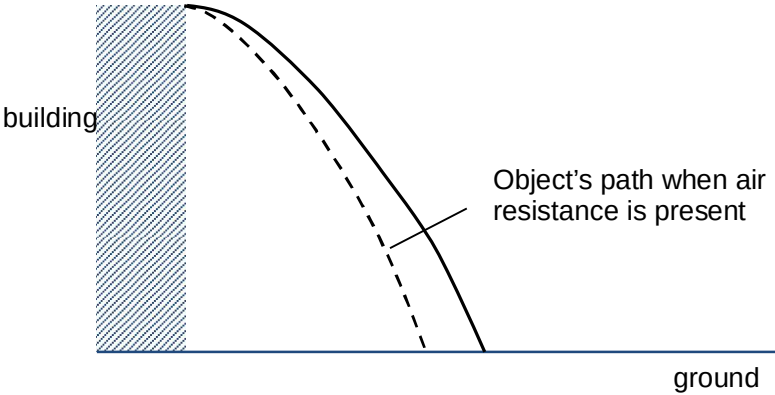


Q.	Solution
1(a)(i)	The object's horizontal component of velocity remains constant in the throughout the motion.
1(a)(ii)	Since air resistance is negligible, the only force acting on the falling object is its weight. The resultant force is the object's weight and is constant. By Newton's 2 nd law, the resultant force gives rise to a constant acceleration. The object's vertical component of velocity will accelerate uniformly from zero throughout the motion.
1(b)(i)	The air resistance will retard the object's horizontal component of velocity as it falls. The horizontal component of velocity will not remain constant but decreases instead.
1(b)(ii)	The resultant force acting on the object is the difference between the object's weight and the air resistance. Since the air resistance is proportional to the object's velocity, resultant force will be reduced as the object falls. By Newton's 2 nd law, the object's acceleration will be reduced and hence the vertical component of velocity will be lesser than in the case of negligible air resistance. When the resultant force reduces to zero, terminal velocity is attained and the object will fall at constant maximum velocity.
1(c)	 The diagram illustrates the effect of air resistance on the path of a falling object. A vertical hatched rectangle on the left is labeled 'building'. A horizontal line at the bottom is labeled 'ground'. Two paths originate from the top edge of the building: a solid curved line and a dashed curved line. Both paths curve downwards and to the right, eventually meeting the ground. The solid line, representing the path with air resistance, is shorter and ends at a point further to the right. The dashed line, representing the path without air resistance, is longer and ends at a point further to the left. A label 'Object's path when air resistance is present' with a pointer indicates the solid line.
2(a)(i)	Radian describes the plane angle subtended by a circular arc as the length of